

Equation Sheet: Magnetism

Intro Physics II • Lectures 18–27 • Knight ch. 29–30

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals.

Symbols and constants

$\vec{B}$ magnetic field (T = N/A·m)	I current (A)	Φ_B magnetic flux (Wb = T·m ²)
q charge (C)	n turns per unit length (1/m)	$\mathcal{E}$ induced emf (V)
$\vec{v}$ velocity (m/s)	N number of turns	$\vec{\mu}$ magnetic dipole moment (A·m ²)
θ angle between the two vectors	r, d distance (m)	μ_0 $4\pi \times 10^{-7}$ T·m/A

1. Magnetic force on charges & currents (Knight 29.1–29.3, 29.7–29.8)

fundamental

Force on a moving charge $\vec{F} = q \vec{v} \times \vec{B}$, $|\vec{F}| = |q| v B \sin \theta$
($\perp$ to both $\vec{v}$ and $\vec{B}$; use the right-hand rule; zero if $\vec{v} \parallel \vec{B}$)

Force on a straight current-carrying wire $\vec{F} = I \vec{L} \times \vec{B}$, $|\vec{F}| = BIL \sin \theta$

derived

Charge in a uniform field ($\vec{v} \perp \vec{B}$) $r = \frac{mv}{|q|B}$
(circular motion; magnetic force does no work, so v is constant)

Period / cyclotron frequency $T = \frac{2\pi m}{|q|B}$, $f = \frac{|q|B}{2\pi m}$
(independent of v and r)

Velocity selector (crossed $\vec{E}$, $\vec{B}$) $v = \frac{E}{B}$

2. Sources of magnetic field (Knight 29.4–29.5)

fundamental

Biot–Savart law — field of a current element $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$
(chop the wire into pieces $d\vec{l}$, add up)

derived

Long straight wire (distance r) $B = \frac{\mu_0 I}{2\pi r}$
(field circles the wire; $B \propto 1/r$)

Center of a circular loop (radius R , N turns) $B = \frac{\mu_0 N I}{2R}$

Inside a long solenoid $B = \mu_0 n I$
(uniform; independent of radius; ≈ 0 outside)

3. Ampère's law (Knight 29.6)

fundamental

Ampère's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$
(any closed loop; I_{enc} is the *signed* current through the loop)

derived

Current entirely outside the loop $\oint \vec{B} \cdot d\vec{l} = 0$

4. Magnetic dipoles & torque (Knight 29.5, 29.9)

fundamental

Dipole moment of a current loop $\vec{\mu} = NI\vec{A}$ ($|\vec{\mu}| = NIA$)
Torque on a dipole in a field $\vec{\tau} = \vec{\mu} \times \vec{B}$, $|\vec{\tau}| = \mu B \sin \theta$
(twists $\vec{\mu}$ toward $\vec{B}$; max at $\theta = 90^\circ$, zero when aligned)

derived

Orientation energy $U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos \theta$
(minimum when $\vec{\mu} \parallel \vec{B}$)
Net force on a dipole in a *uniform* field $\vec{F}_{\text{net}} = 0$ (torque only)

5. Magnetic flux & electromagnetic induction (Knight 30.1–30.7)

fundamental

Magnetic flux through a loop $\Phi_B = \int \vec{B} \cdot d\vec{A}$ $\xrightarrow{\text{uniform}}$ $\Phi_B = BA \cos \theta$
Faraday's law — induced emf $\mathcal{E} = -N \frac{d\Phi_B}{dt}$
(only a *changing* flux induces an emf)
Lenz's law induced current opposes the change in flux

derived

Motional emf (bar of length L , speed v , $\vec{v} \perp \vec{B}$) $\mathcal{E} = BLv$
Induced current and dissipated power $I = \frac{\mathcal{E}}{R}$, $P = \frac{\mathcal{E}^2}{R} = F_{\text{pull}} v$

Companion simulations at hoppese.github.io/Intro-physics-sims/: magnetic-field-of-currents, discrete-current-builder-3d, ampere-loop, dipole-field-and-torque, lorentz-force, current-loop-motor, faraday-induction, faraday-flux-explorer, rail-flux-bar.